

Andrey Zakharov

ML Engineer

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ML Engineer with 3+ years of hands-on project-based development experience. Builds applied machine-learning and computer-vision systems across structured data, multimodal sensor data and image-processing pipelines. Experienced in data preparation, feature engineering, model training and evaluation, inference optimization and integrating ML models into working products.

Experience

Independent Project Development: 2023 – Present.

COMPASS AI – Applied ML Engineer

2026 – Present

Tech: Python, PyTorch, scikit-learn, Pandas, NumPy, ONNX, ONNX Runtime, sentence-transformers, FastAPI

- Designed an end-to-end ML pipeline for task-to-employee matching with task, employee and pair-level features, group-based splitting by task ID and zero leakage across train, validation and test sets.
- Compared six approaches, including rule-based matching, classical ML models and a PyTorch MLP; achieved ROC-AUC 0.859, MRR 0.944 and P@1 0.893 on 375 held-out tasks.
- Built 384-dimensional multilingual task-text embeddings and feature schemas with 404 task, 60 employee and 13 pair-level features across 42,052 training rows and 8,924 test rows.
- Exported the PyTorch model to ONNX opset 17 and validated inference parity with maximum absolute difference of 2.38e-7; added fairness and explainability reports for recommendations.

Russian Automotive Assistance System (RAAS) – Computer Vision & ML Engineer

2023 – 2024

Tech: Python, PyTorch, torchvision, OpenCV, NumPy, CARLA, CUDA, SQLite

- Collected and prepared a multimodal smart-parking dataset with 13,000+ camera images, synchronized driving telemetry and four radar channels; data was captured from four cameras and recorded every fifth frame.
- Trained a four-camera MultiCamNet based on pretrained ResNet34 branches, concatenating visual features with radar measurements to predict throttle, steering and braking commands.
- Implemented training with augmentation, normalization, SmoothL1 loss, Adam optimization, cosine annealing, validation split and early stopping; built inference with calibrated control-output multipliers.
- Integrated RGB cameras, LiDAR and radar into CARLA-based simulation scenarios for lane detection, obstacle analysis, emergency braking and vehicle-control experiments.

Face-Swap-Studio – Applied Computer Vision Engineer

2025

Tech: Python, InsightFace, ONNX Runtime, OpenCV, NumPy, PyTorch, FastAPI

- Integrated nine face-swap and head-model backends through a common registry with model-readiness checks, isolated runners and metadata for multi-face and per-face assignment support.
- Built image-processing and inference workflows for face detection, bounding-box validation, cropping, target-face mapping, single-image processing and batch generation.
- Added optional GFPGAN restoration and Real-ESRGAN upscaling, temporary-data isolation, model status checks and Apple Silicon/MPS-compatible local execution.

UrbanFlow AI – Simulation & Applied ML Engineer

2026

Tech: Python, SUMO, TraCI, Pandas, Jupyter, FastAPI, JSON/JSONL

- Designed and evaluated a policy-based traffic-light controller using lane queues, halted vehicles, waiting time, speed, occupancy and phase-pressure signals with hold/switch actions and timing constraints.
- Built an experiment pipeline with curriculum configuration, saved checkpoints, model registry, CSV/JSONL histories, Markdown dashboards and Jupyter analytics for simulation runs with 500 vehicles and up to 2,241 controlled traffic lights.
- Integrated the controller with SUMO and TraCI to evaluate traffic-management policies in reproducible microscopic simulation scenarios.

Botyan – Applied NLP/LLM Engineer

2023 – 2024

Tech: Python, asyncio, aiogram, discord.py, OpenRouter/OpenAI-compatible APIs

- Implemented contextual LLM workflows for Telegram and Discord using shared conversation history, up to 100 messages for chat context and up to 1,000 messages for image-caption generation.

Skills

Technical: Python, PyTorch, scikit-learn, torchvision, OpenCV, Pandas, NumPy, ONNX/ONNX Runtime, InsightFace, Sentence Transformers, CUDA, CARLA, SUMO/TraCI, Jupyter, FastAPI, SQLite, Git.

Languages: English – C1 (Advanced); Russian – Native.

Education

SAINT PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (SUAI) – Saint Petersburg, Russia

- Master's Degree, Management – Strategic Human Resource Management 2027
- Bachelor's Degree, Software Engineering – Software Systems Design 2025
- Professional Retraining Diploma – Web and Multimedia Applications Developer 2024
- Additional Professional Education: 250+ certificates and diplomas from 60+ universities, companies, educational platforms, institutes, and professional training programs worldwide, covering diverse academic and professional fields and disciplines.